

ASHUTOSH MEENA

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SUMMARY

Computer Engineering Graduate with experience in Java-based microservices, web application development, and cloud infrastructure. Proficient in Spring Boot, REST APIs, Docker, Kubernetes (AKS), Kafka, and Elasticsearch. Built and deployed scalable applications with automated testing and CI/CD workflows. Developed full-stack IoT and ML solutions using Python and ReactJS. Strong understanding of cloud-native architecture, container orchestration, and system performance optimization. Seeking roles in web development, Java backend, or cloud administration.

EDUCATION

Illinois Institute of Technology, Chicago, IL

May 2025

Masters of Science, Computer Engineering, 3.62

Relevant course work: Object Oriented Programming, Cloud Computing, Application Software Design, Cyber Security, Machine and Deep Learning, Secure Machine Learning, AI in Smart Grid

Dr. A. P. J. Abdul Kalam Technical University, Lucknow, UP, India

September 2020

Bachelors of Technology, Electronics and Communication Engineering, 3.06

WORK EXPERIENCE

System Engineer

Infosys Ltd, Bengaluru, India

February 2021 - October 2022

- Migrated web application from PCS - cassandra to Azure Kubernetes Service (AKS) - elasticsearch improving performance by **30%** and scalability by **45%**, also conducted load testing with Apache J-meter and resolved issues to reduce error rate to below **1%**.
- Constructed and upgraded **18** micro-services based on JAVA-SpringBoot-Maven interfacing with elasticsearch and Apache-Kafka, regularly deploying in production serving more than 150 client applications.
- Engaged in daily scrum calls as well as weekly team-building exercises, fostering a collaborative work environment and strengthening interpersonal relationships among colleagues.

PROJECTS

Energy Load Forecasting Using Machine Learning

March 2025 - May 2025

- Developed and evaluated forecasting models (**Linear Regression** and **MLPRegressor**) to predict hourly electrical load across 20 zones using weather and historical load data.
- Applied feature engineering techniques and Pearson correlation analysis to map relevant weather stations to load zones, improving model relevance and reducing data dimensionality.
- Achieved high predictive accuracy (**$R^2 = 0.954$** , **MAPE = 0.306**) using optimized MLPRegressor with GridSearchCV and early stopping, outperforming baseline models.

Wapp Controlled Pi-Camera

June 2024 - July 2024

- Developed an application for Raspberry-pi in python leveraging Selenium to automate whatsapp log-in and messaging.
- Integrated OOP-Design in the application containing 7 separate python based microservices for different feature applications.
- Configured night vision camera control for real-time monitoring and observation with user controle over WhatsApp.
- Designed device to read and execute WhatsApp commands, capturing images, recording videos, and performing 0 to 180-degree servo rotations.

IoT Smart Plug Simulator

February 2024 - May 2024

- Developed a springboot-gradle web application for controlling a smart plug with more than 75 unit test cases.
- Implemented application utilizes MQTT network protocol and functionalities involved controlling a single plug or multiple plugs as a group.
- Learned and applied ReactJS for user-interface development for web-application with 95% accuracy in terms of unit test cases

Scalable Microservices Deployment with Docker, K8s & Chaos Engineering

September 2023 - December 2023

- Developed containerized microservices using Docker, Kubernetes, and Cassandra with automated orchestration and cluster management.
- Configured and tested a 4+ node Cassandra cluster via Docker Compose, enhancing performance and fault tolerance.
- Optimized a Kafka-based real-time data streaming system on Kubernetes, enhancing fault tolerance and high availability.
- Executed Chaos Engineering experiments with Chaos Mesh, maintaining system resilience despite 50% pod failures.

SKILLS

Programming Languages: C | C++ | JAVA | Python

Web Development: Spring Boot | REST- API | Maven | Gradle

Cloud-Native Services: Kafka | Kafka-Streams (JAVA) | Azure-Kubernetes | Docker

Database - Elasticsearch DB | Cassandra DB | PostgreSQL

Machine Learning - Regression | Classification | Neural Network | Convolutional Neural Network | Deep Neural Network

ML/AI Tools : PyTorch | TensorFlow | TensorFlow-Lite | Pg-Vector

Other Tools - GitHub | Azure-DevOps | Linux | Postman | J-Unit | Selenium